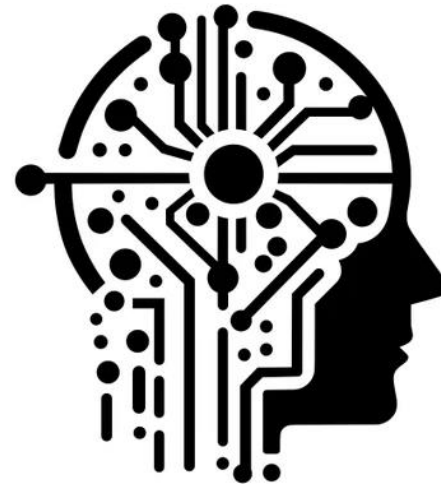


Bringing Reasoning to Artificial Intelligence Networks (BRAIN)

2nd year Research & Development Project



Paul Lefrais, Selen Erden

Project Supervisor: Sebastien Schwartz

Bringing Reasoning to Artificial Intelligence Networks (BRAIN)

AGI: Artificial General Intelligence



Presentation overview

1. Presentation

Current context in the AI field

2. BRAIN

Presentation of the BRAIN approach

3. Knowledge creation

Development of initial knowledge creation

4. Results

Results and discussion

1. Presentation

2. BRAIN

3. Knowledge creation

4. Results

Current leading LLMs



LLM : Large Language Model

ARC AGI Benchmark



François Chollet

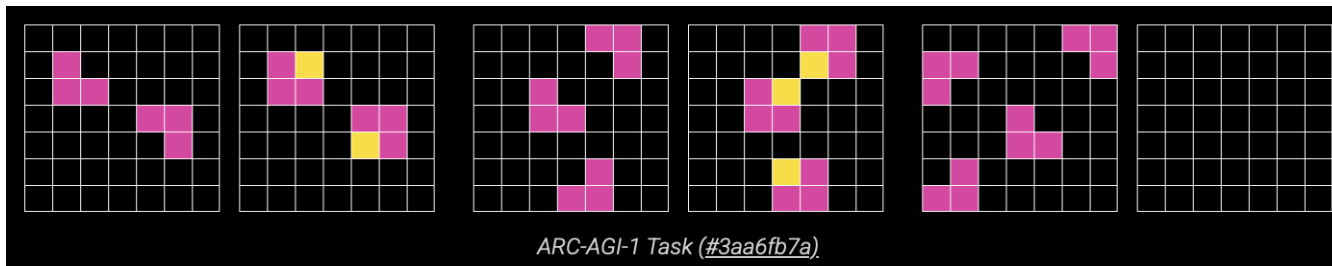
1. Presentation

2. BRAIN

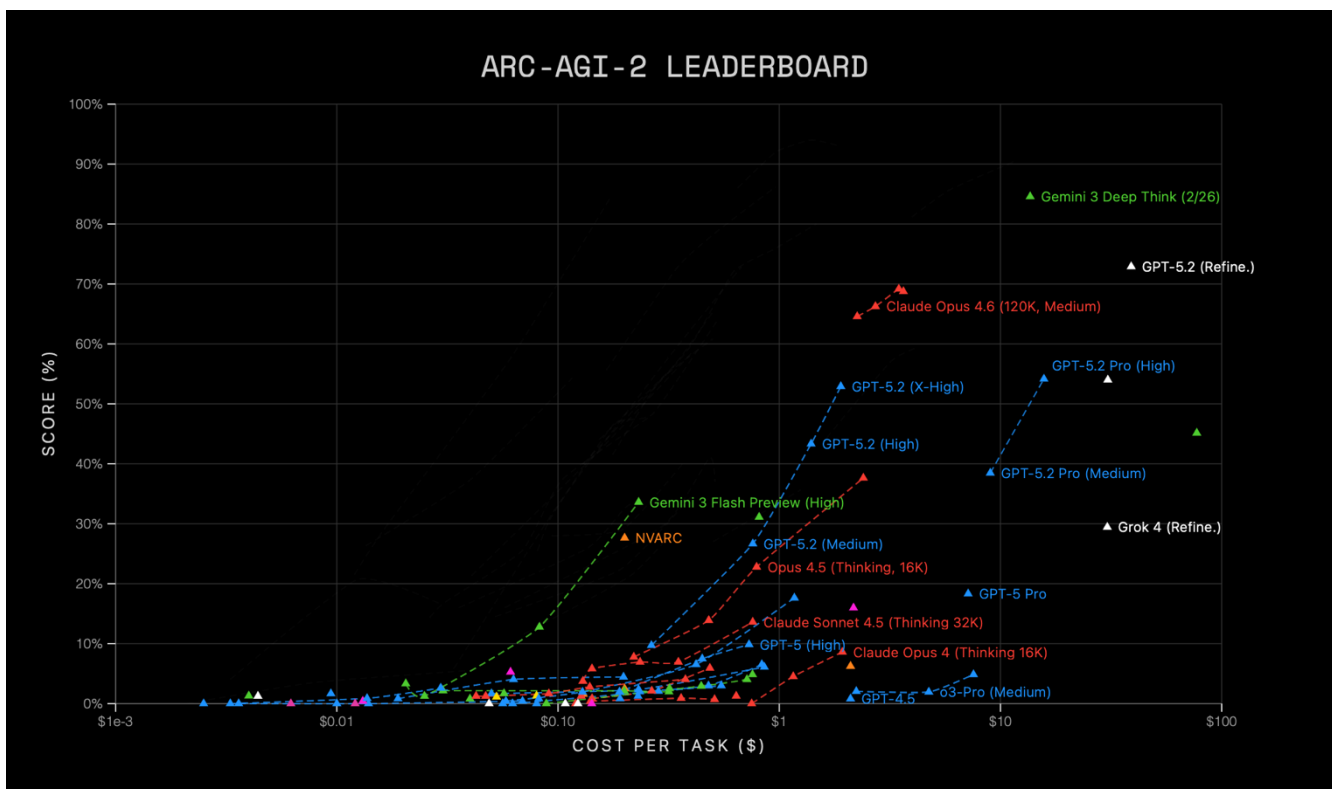
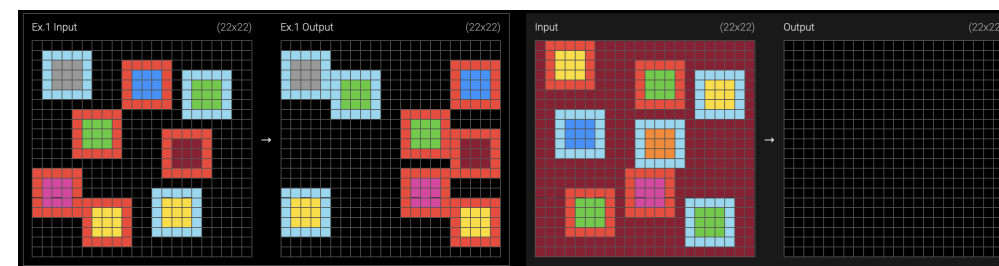
3. Knowledge creation

4. Results

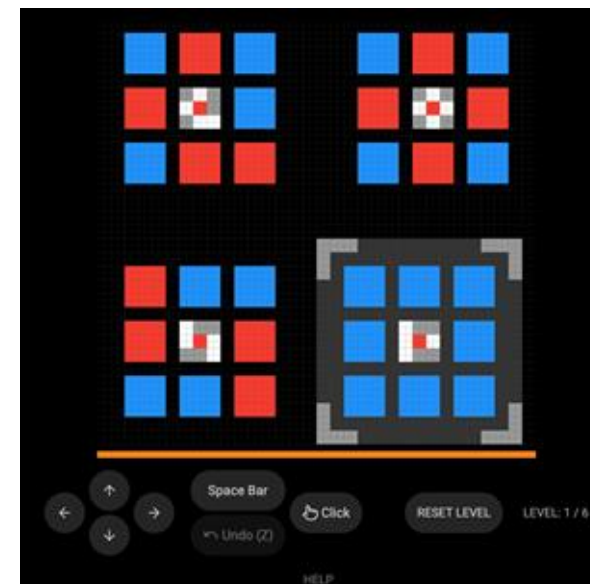
ARC AGI Benchmark V1



V2



V3 (coming soon)



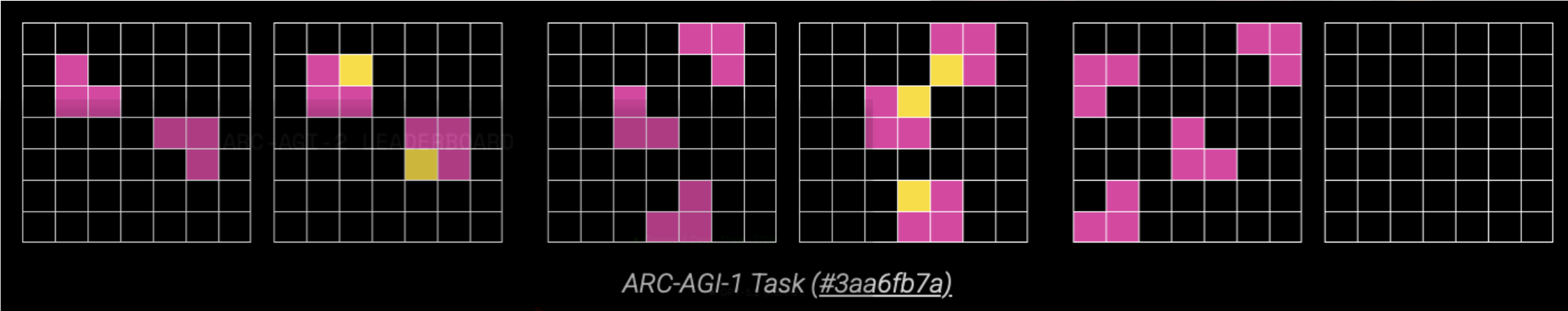
1. Presentation

2. BRAIN

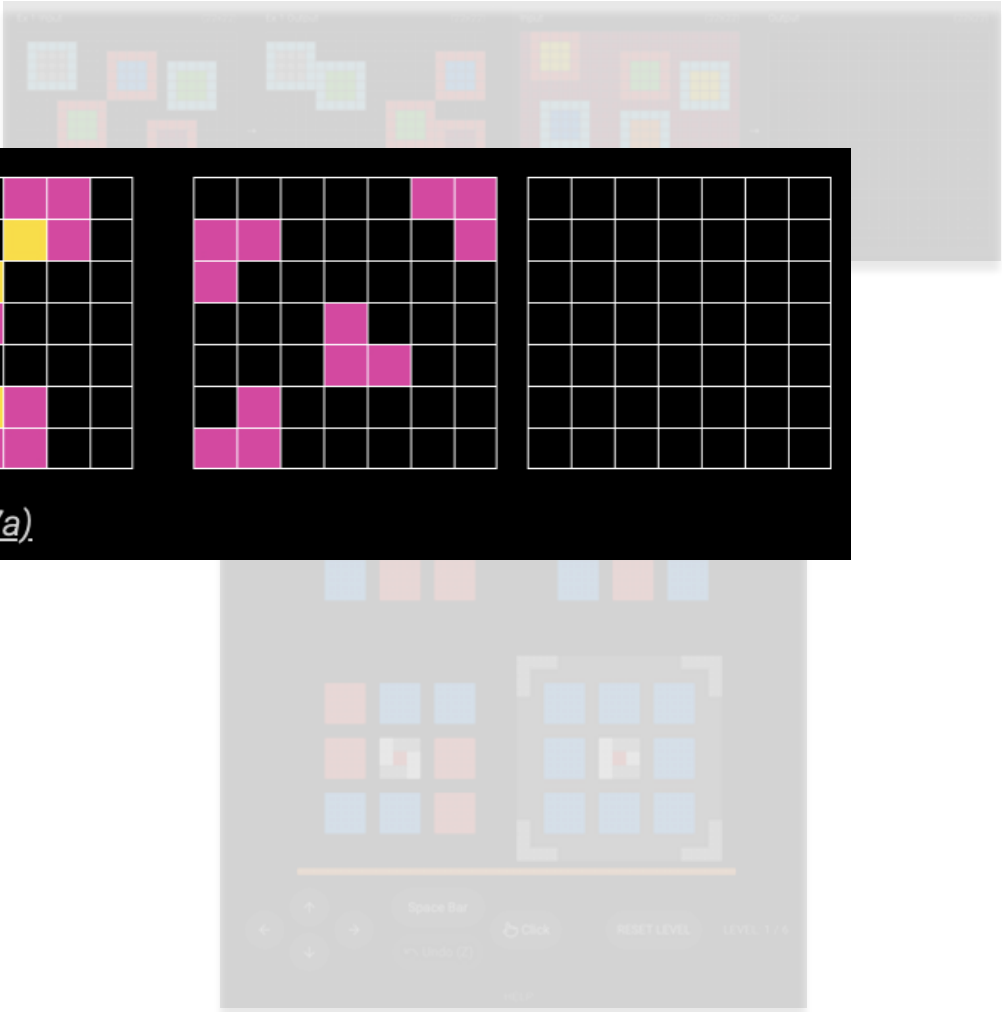
3. Knowledge creation

4. Results

ARC AGI Benchmark V1



ARC-AGI-1 Task ([#3aa6fb7a](#))



1. Presentation

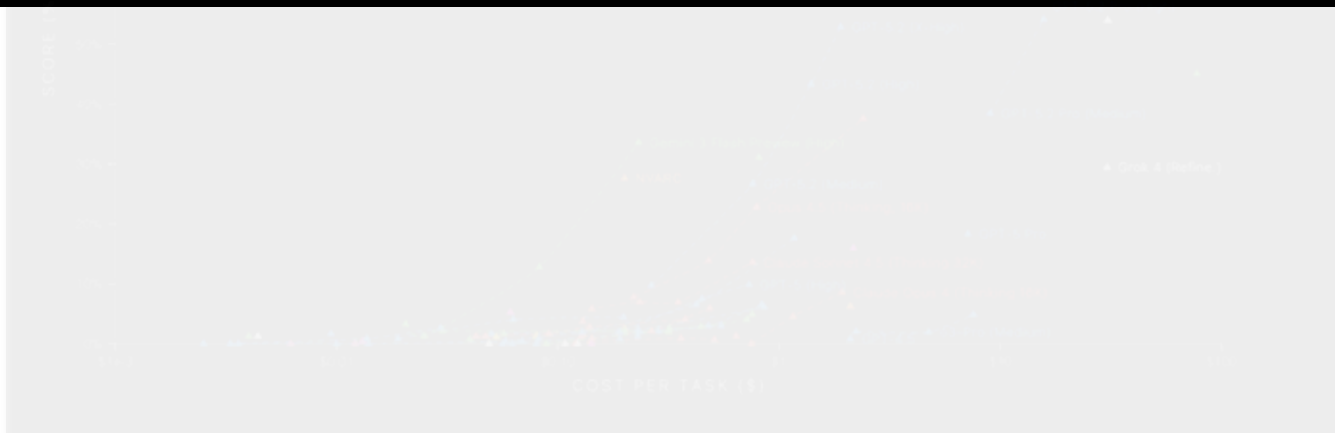
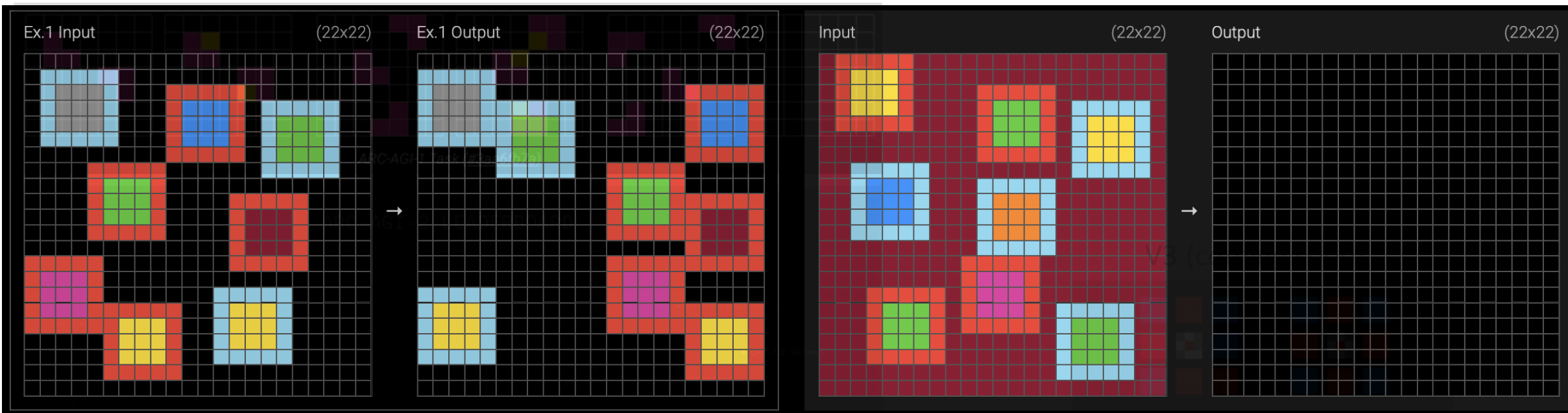
2. BRAIN

3. Knowledge creation

4. Results

ARC AGI Benchmark V1

2nd Version



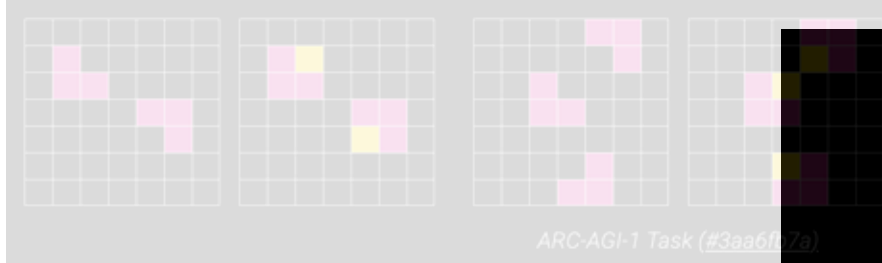
1. Presentation

2. BRAIN

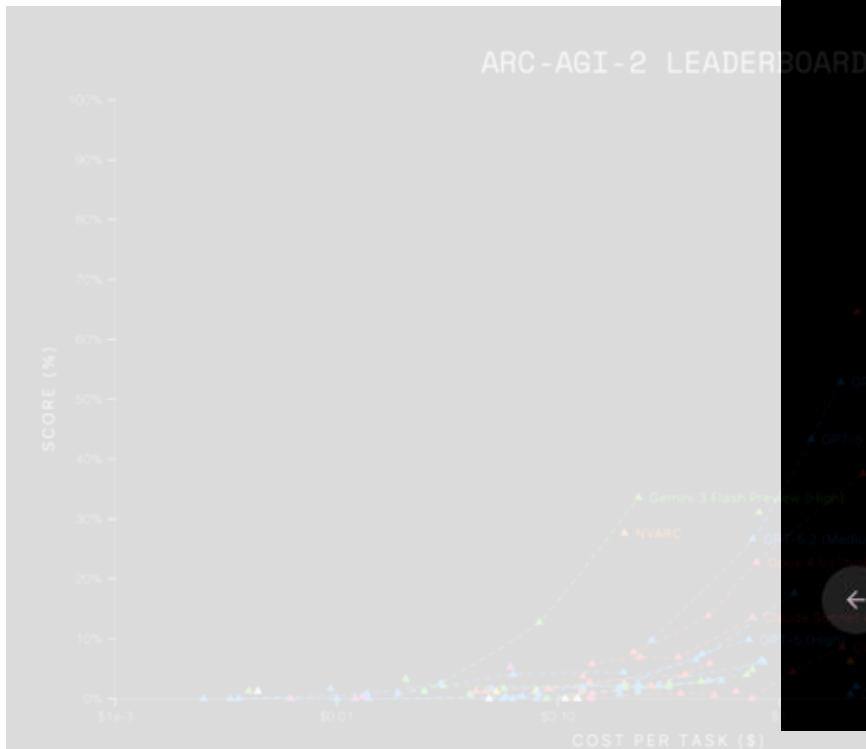
3. Knowledge creation

4. Results

ARC AGI Benchmark V1

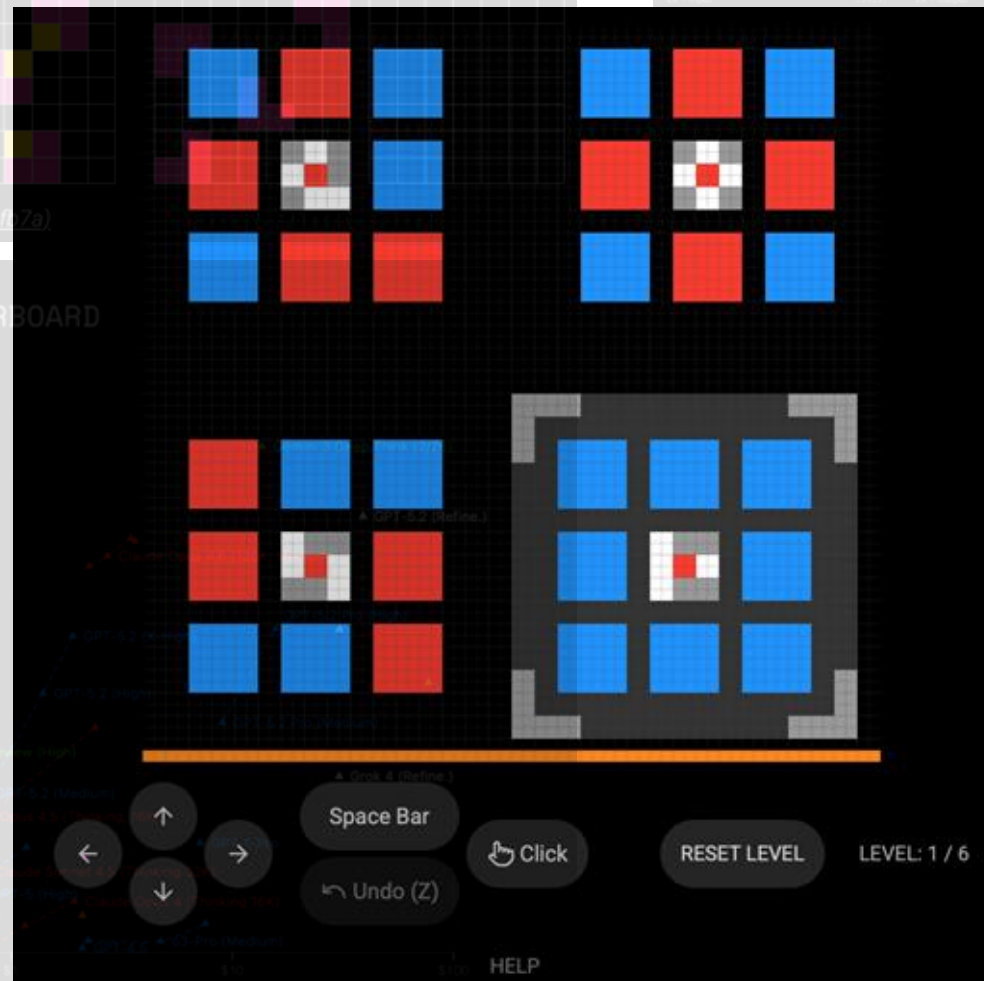


ARC-AGI-1 Task (#3aa6f7a)



V3 (coming soon)

V2

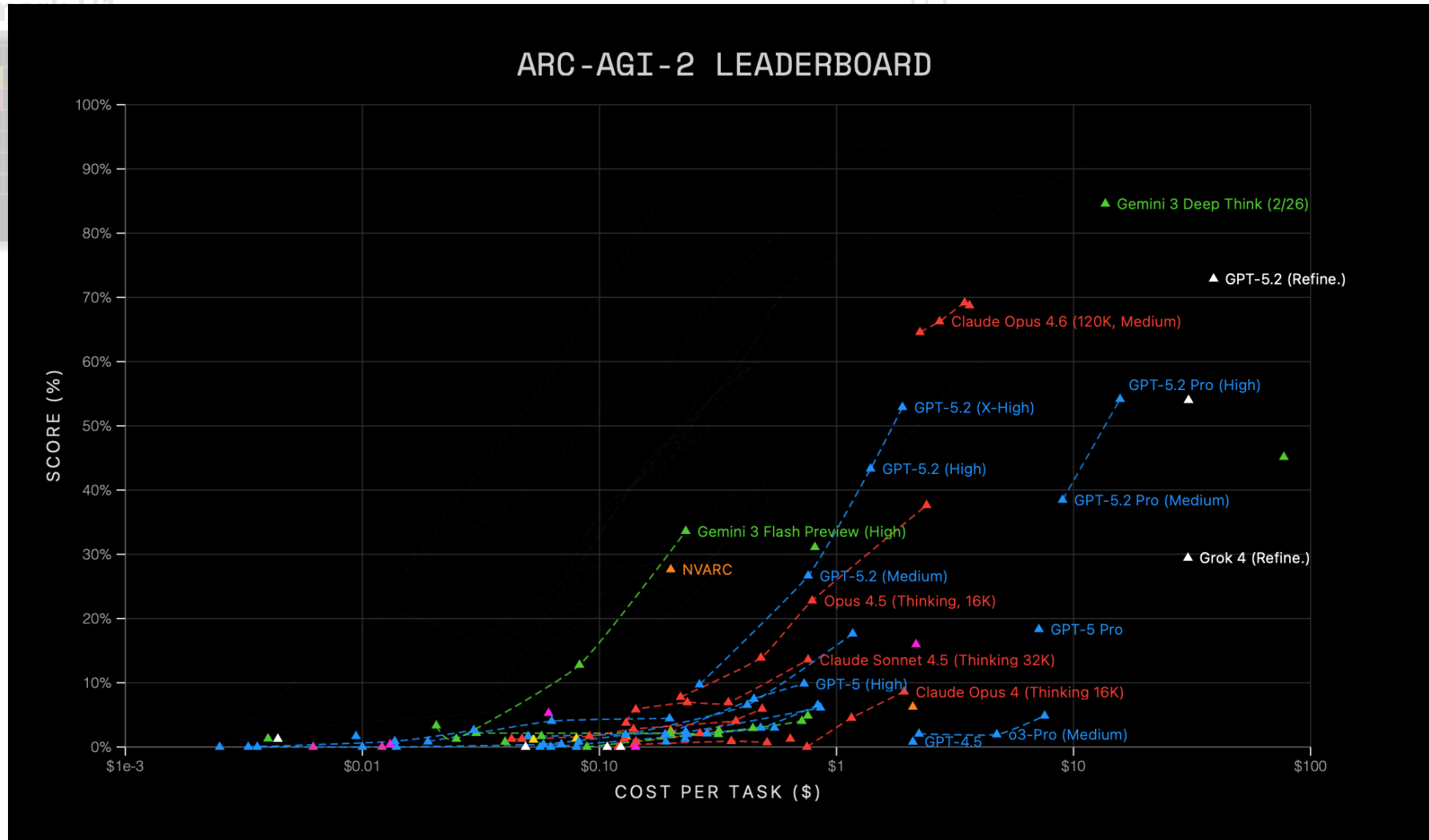


1. Presentation

2. BRAIN

3. Knowledge creation

4. Results



1. Presentation

2. BRAIN

3. Knowledge creation

4. Results

1. Presentation

2. BRAIN

3. Knowledge creation

4. Results

1

Knowledge creation

2

Action creation

1. Presentation

2. BRAIN

3. Knowledge creation

4. Results

Knowledge creation

Action creation

1. Presentation

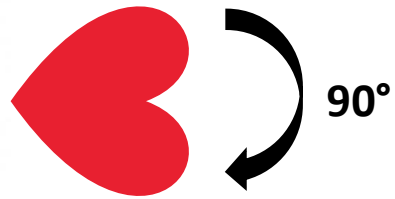
2. BRAIN

3. Knowledge creation

4. Results

Knowledge creation

Action creation



Analogy of a children's game

1. Presentation

2. BRAIN

3. Knowledge creation

4. Results

Knowledge creation

Action creation

Lines / Points

Usual shapes

Form filling

Translations & Rotations



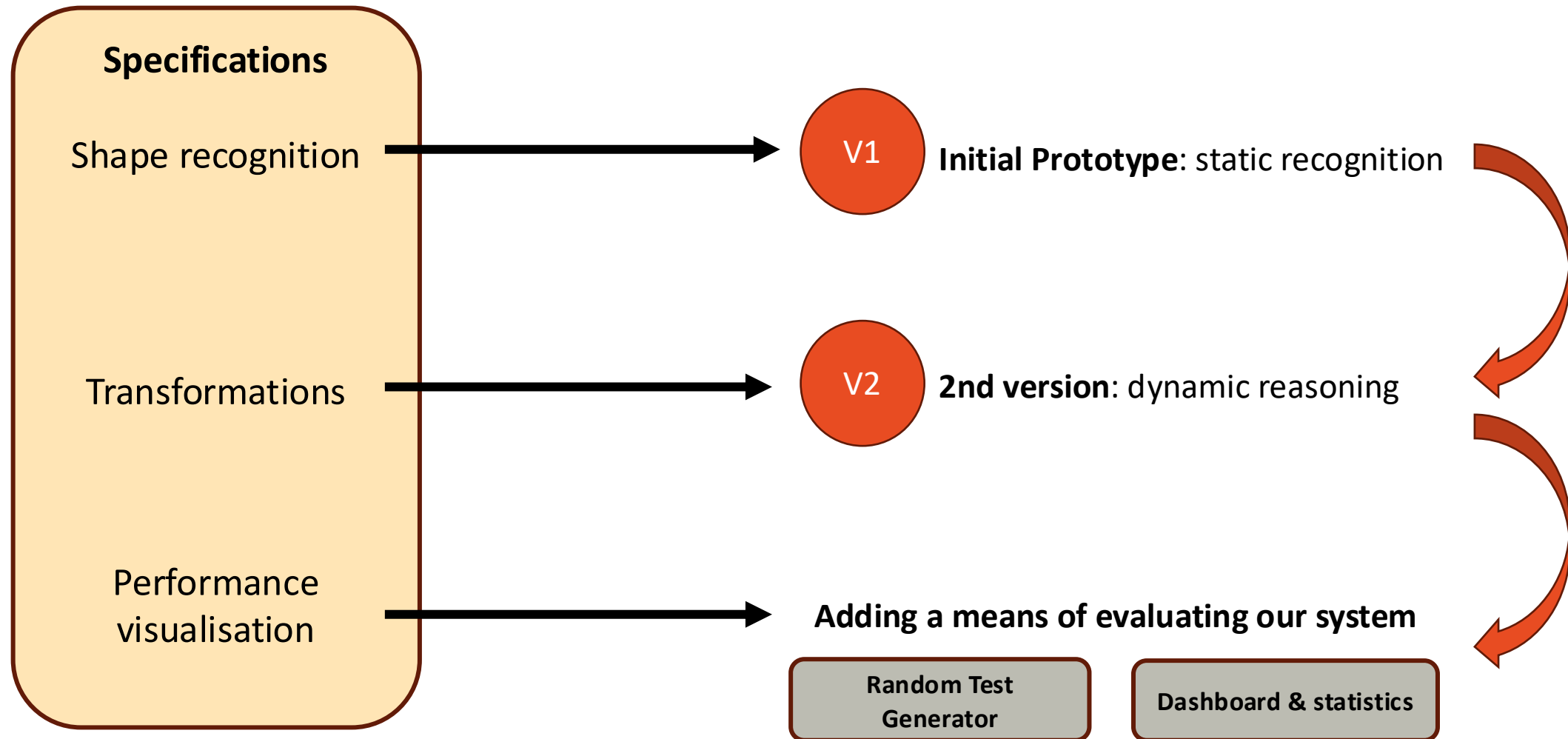
Pattern recognition and
basic transformations

entrance grid
<raw shapes>

BRAIN system

Logical comprehension
of transformation rules

Constructive approach



1. Presentation

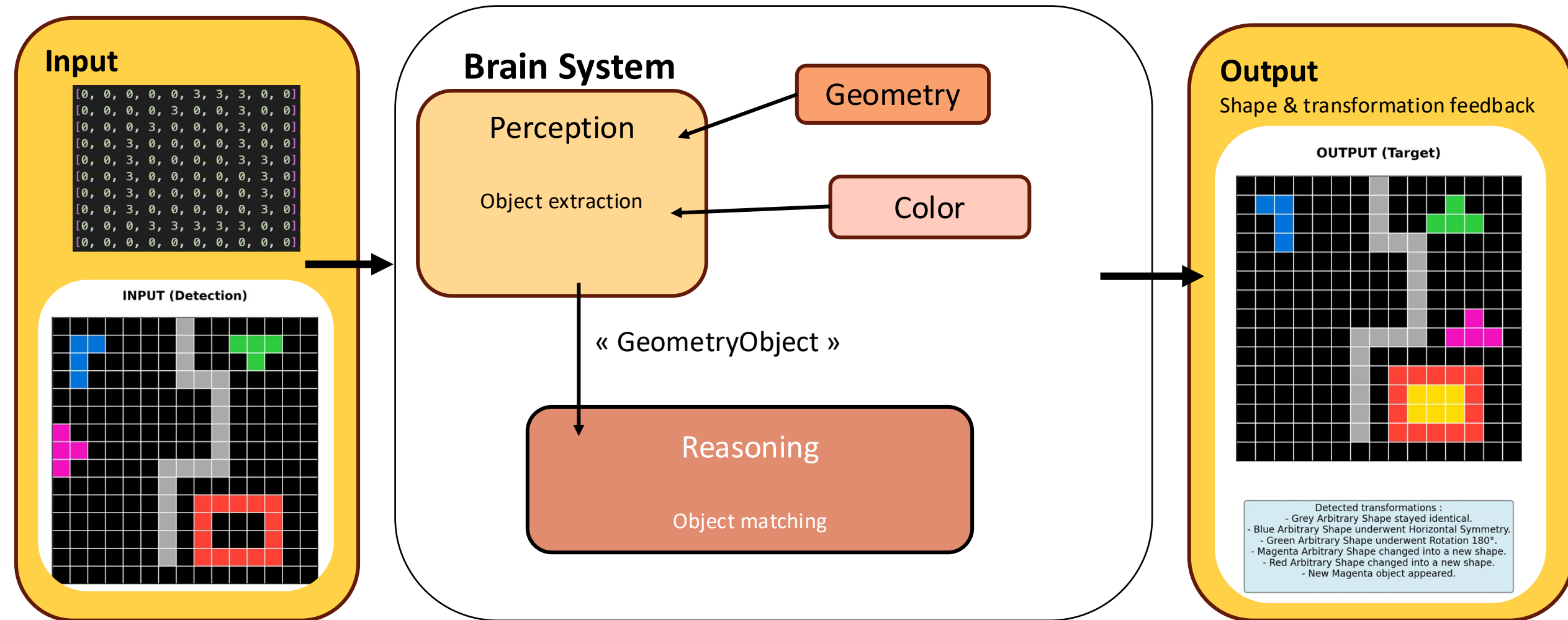
2. BRAIN

**3. Knowledge
creation**

4. Results

System architecture: overall diagram

System architecture: overall diagram



1. Presentation

2. BRAIN

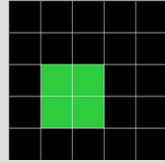
3. Knowledge
creation

4. Results

Overall system architecture

Geometry

Basic geometric concepts



```
40 |  
41 |     self.shape = "point"  
42 | elif w == 1 or h == 1:  
43 |     self.shape = "line"  
44 | elif area == bounding_box_area:  
45 |     self.shape = "square" if w == h else "rectangle"  
46 | else:  
47 |     self.shape = "other"
```

Color

To match with
the ARC colors

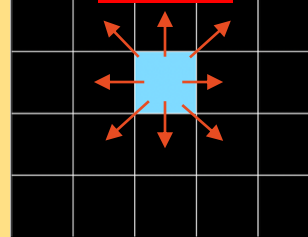
```
COLOR_HEX = [  
    '#000000', # 0 Black  
    '#0074D9', # 1 Blue  
    '#FF4136', # 2 Red  
    '#2ECC40', # 3 Green  
    '#FFDC00', # 4 Yellow  
    '#AAAAAA', # 5 Grey  
    '#F012BE', # 6 Magenta  
    '#FF851B', # 7 Orange  
    '#7FDBFF', # 8 Cyan  
    '#870C25', # 9 Brown  
]
```

Perception

```
grid_in = [  
    [3, 3, 0, 0, 0],  
    [3, 3, 0, 0, 0],  
    [0, 0, 0, 0, 0],  
    [0, 0, 0, 3, 3],  
    [0, 0, 0, 3, 3],  
]
```

Detected objects &
transformation list

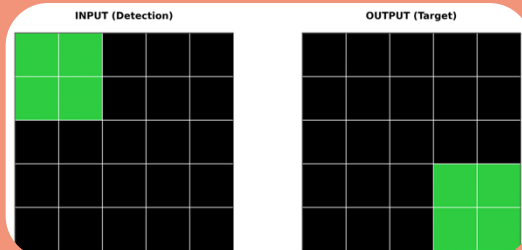
8-connectivity
analysis



Creation of a « **GeometryObject** »
with adjoining pixels

Reasoning

Comparison



Object matching

Transformation
detection

Output :
Semantic
explanation in
natural language

1. Presentation

2. BRAIN

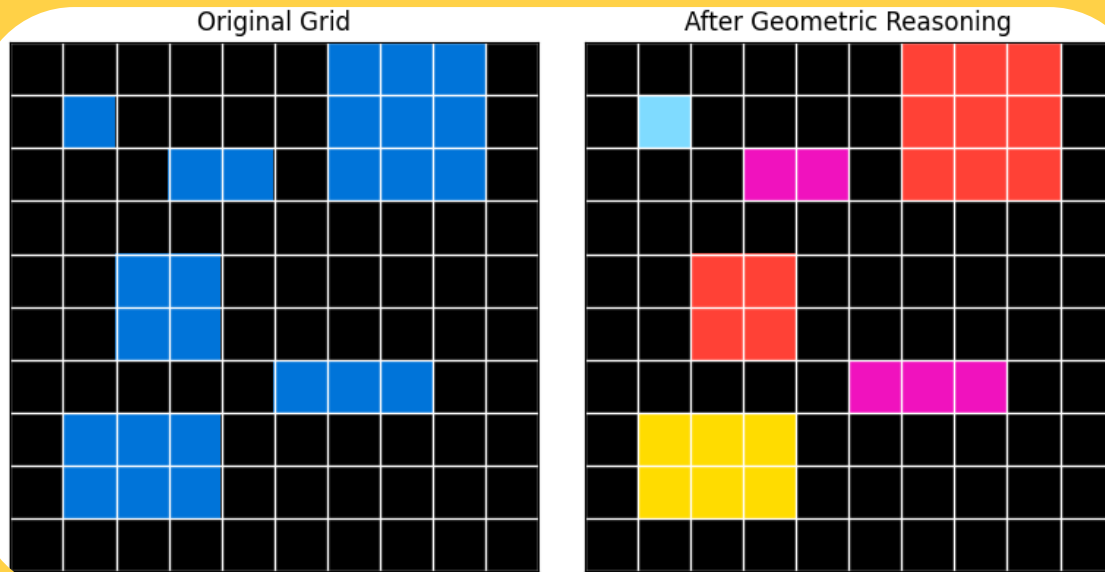
3. Knowledge creation

4. Results

Phase 1: Initial prototype

Phase 1: Initial prototype

Form detection - example

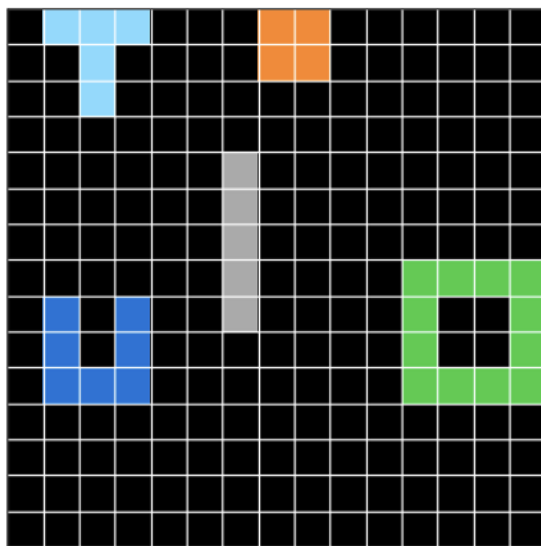


BRAIN V1: Static shape identification

- Shape detection
- Color code for:
 - Points
 - Lines
 - Squares
 - Rectangles

Phase 2: Dynamic Reasoning

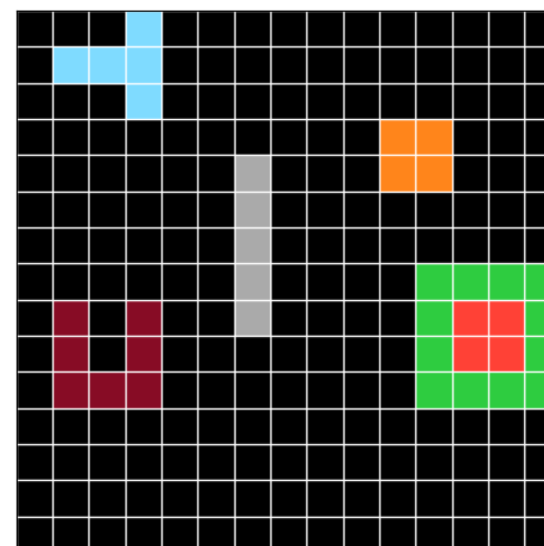
INPUT (Detection)



Objects detected :

- other (Cyan)
- square (Orange)
- line (Grey)
- other (Green)
- other (Blue)

OUTPUT (Target)

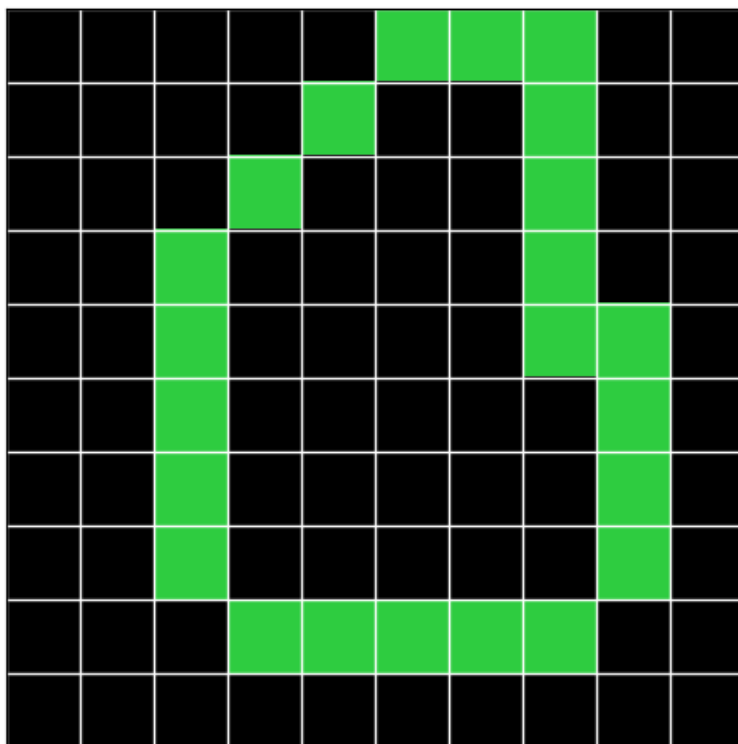


Detected transformations :

- Azure Arbitrary Shape underwent Rotation 270°.
- Orange Square underwent Translation (+3, +3).
- Grey Line stayed identical.
- Green Arbitrary Shape stayed identical.
- Blue Arbitrary Shape became Maroon (Color).
- Green object was filled with Red (Fill).

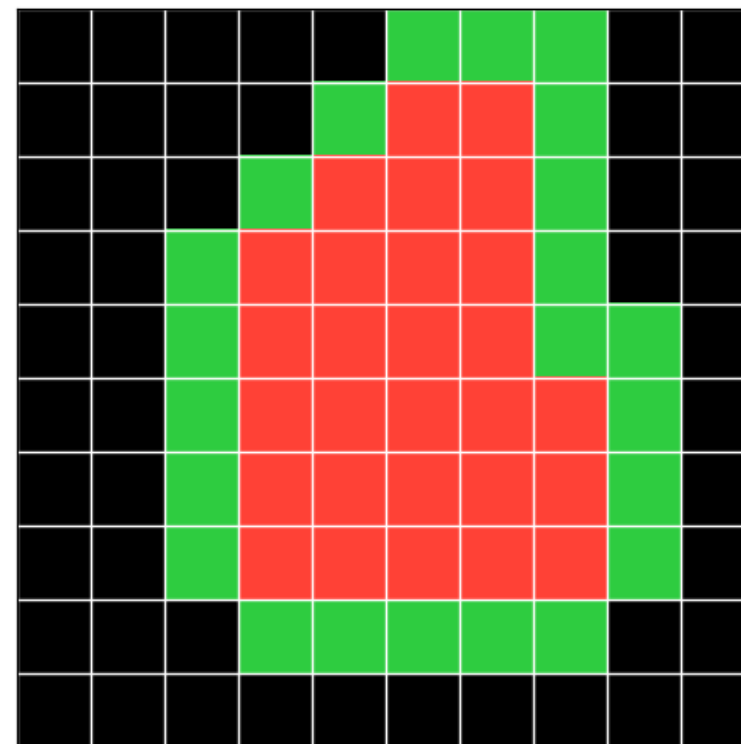
Phase 2: Dynamic Reasoning

INPUT (Detection)



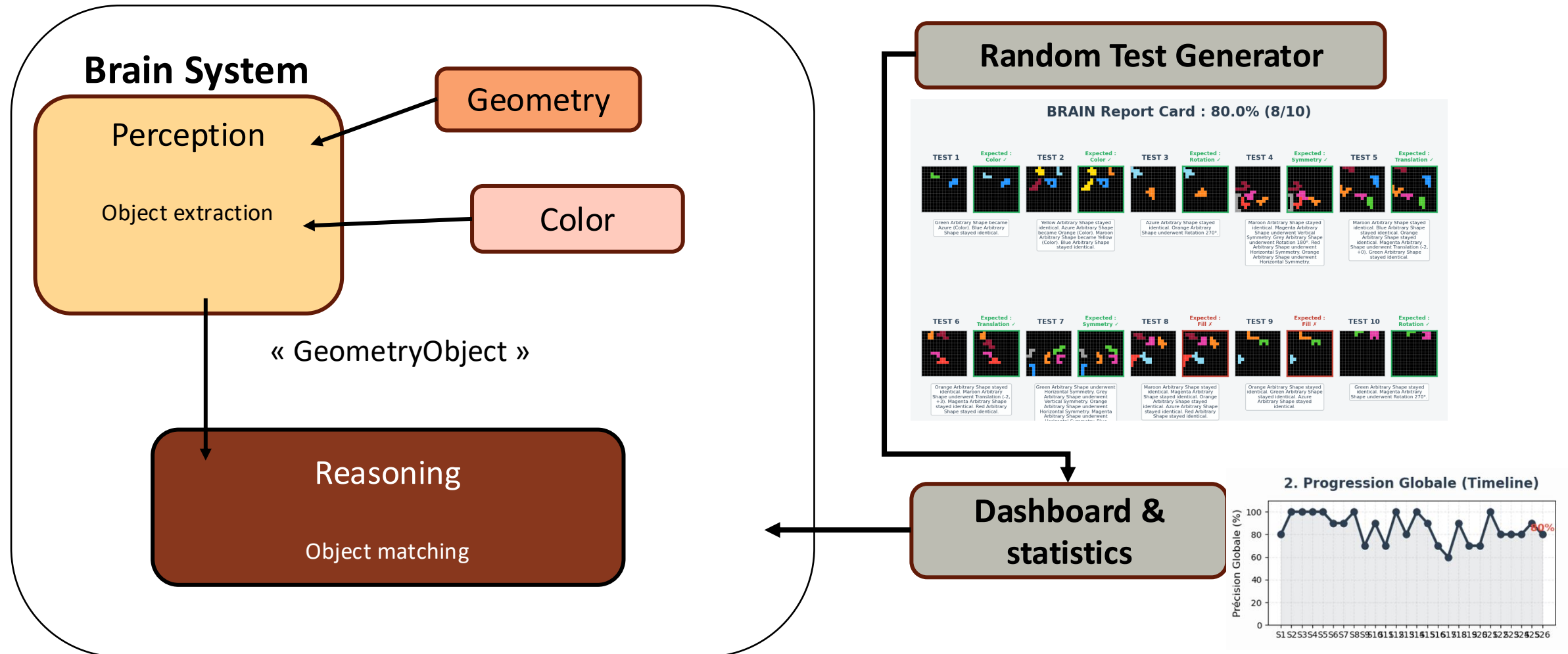
Objects detected :
- other (Green)

OUTPUT (Target)



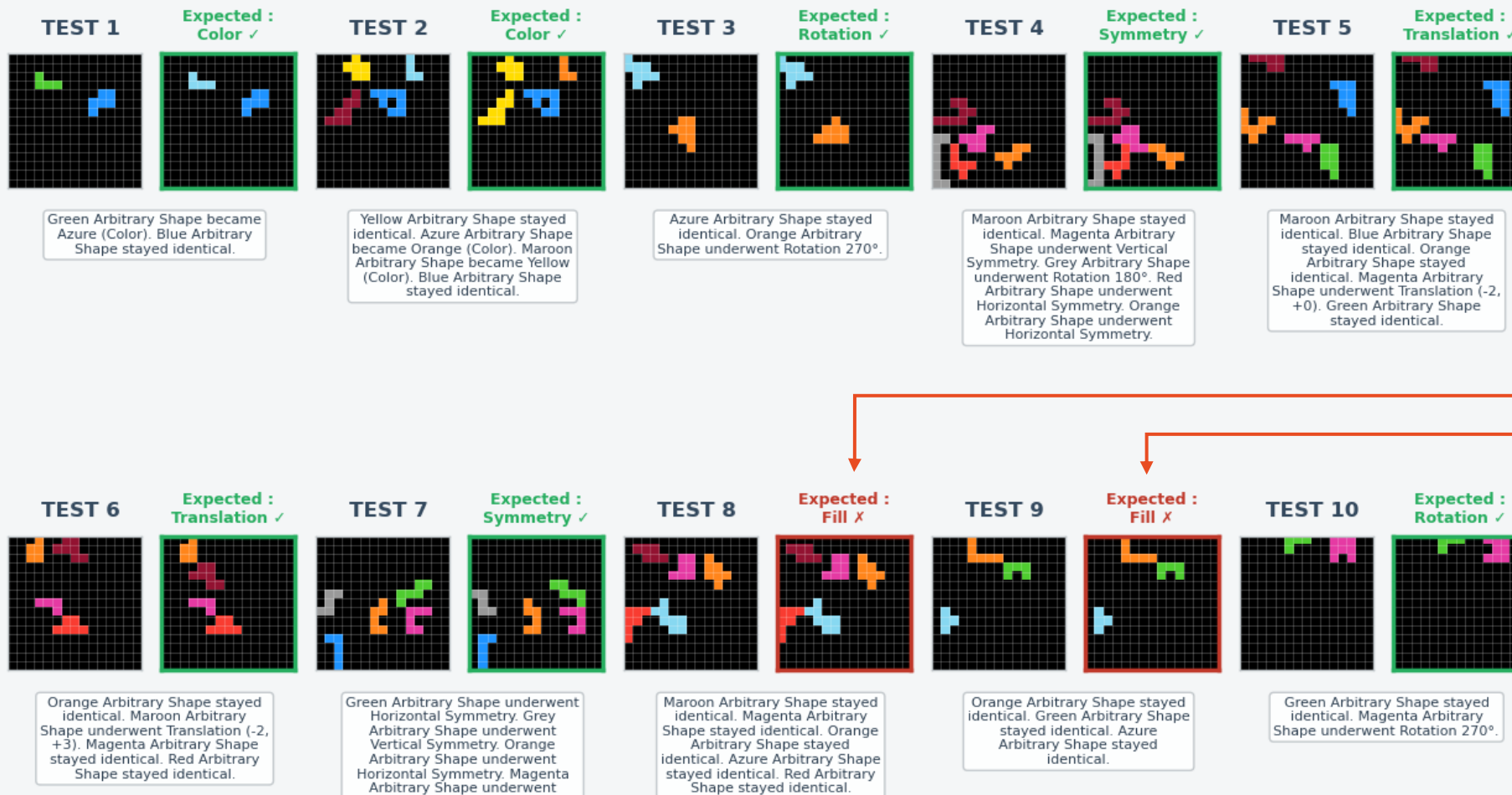
Detected transformations :
- Green Arbitrary Shape stayed identical.
- Green object was filled with Red (Fill).

Creation of a Random Test Generator and Dashboard



Random Test Generator

BRAIN Report Card : 80.0% (8/10)



→ System evaluation tool

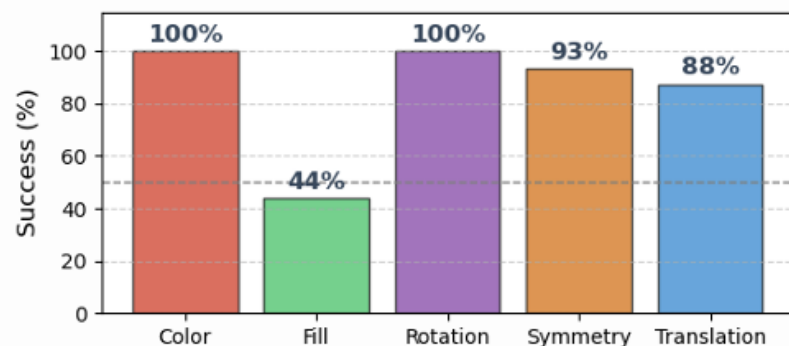
→ Reveals potential errors



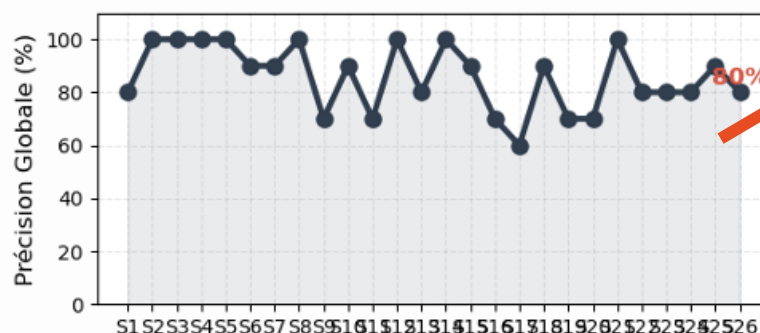
Some limits regarding the generator

Dashboard

1. Success Rate by Rule



2. Progression Globale (Timeline)



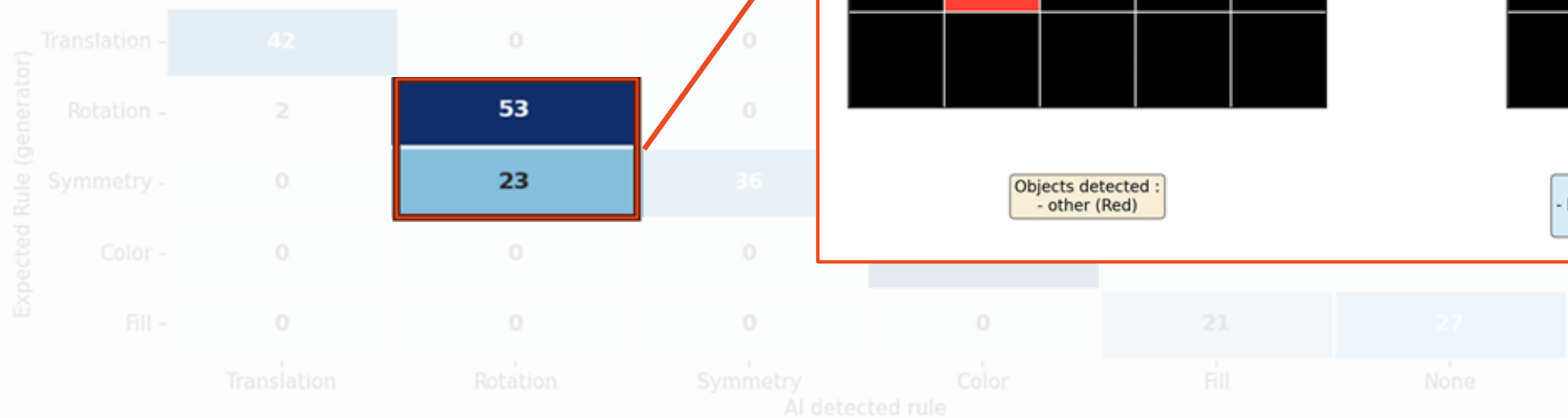
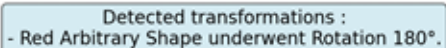
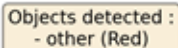
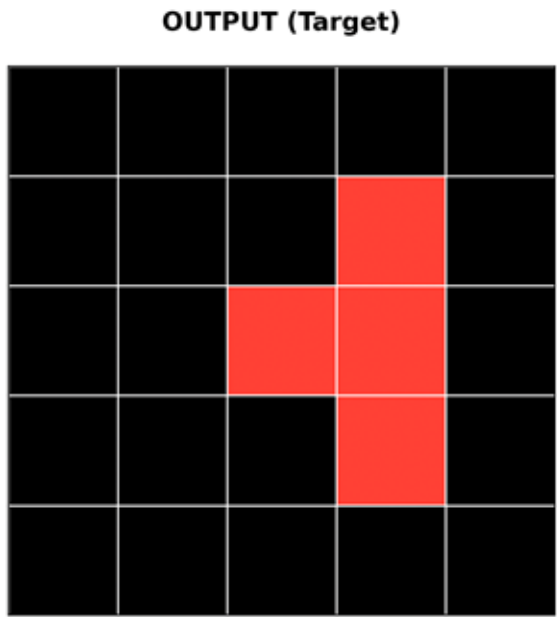
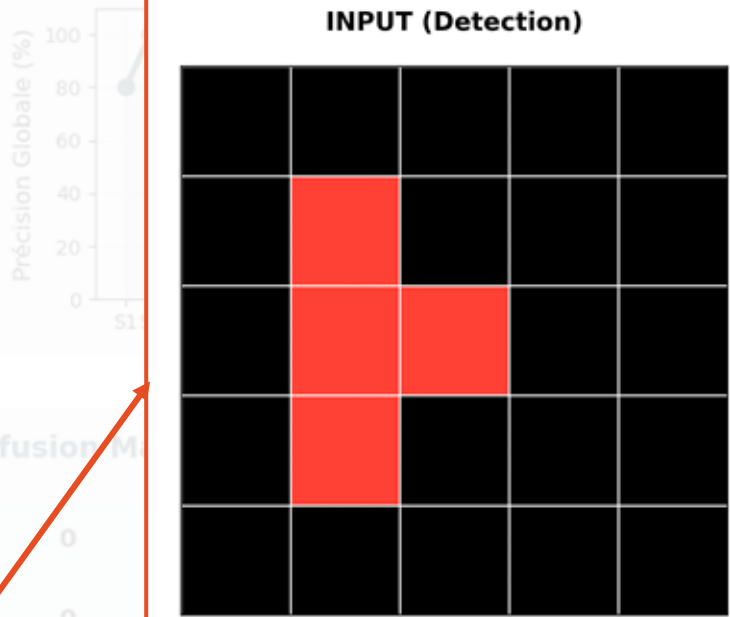
Reason of the variations:

- random nature of the generated shapes

3. Confusion Matrix (Error Diagnosis)

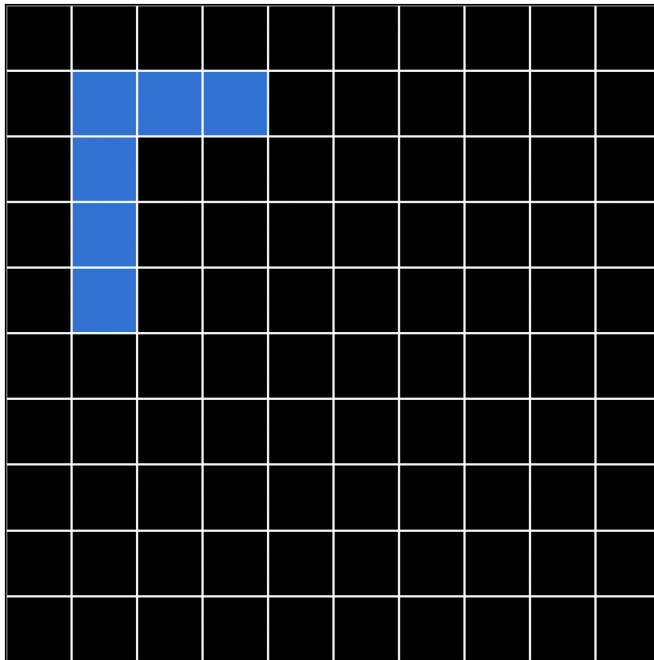
Expected Rule (generator)	Translation	Rotation	Symmetry	Color	Fill	None
	Translation	42	0	0	0	6
	Rotation	2	53	0	0	0
	Symmetry	0	23	36	0	0
	Color	0	0	0	50	0
AI detected rule	Fill	0	0	0	21	27
	None	0	0	0	0	0

Symmetry vs Rotation



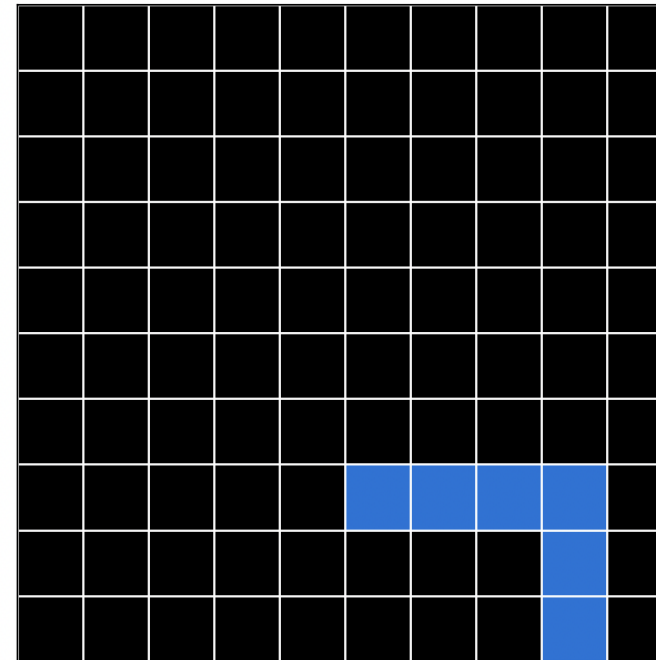
Challenge for the next phase of the project

INPUT (Detection)



Objects detected :
- other (Blue)

OUTPUT (Target)



Detected transformations :
- Blue Arbitrary Shape underwent Rotation 270°.

1. Presentation

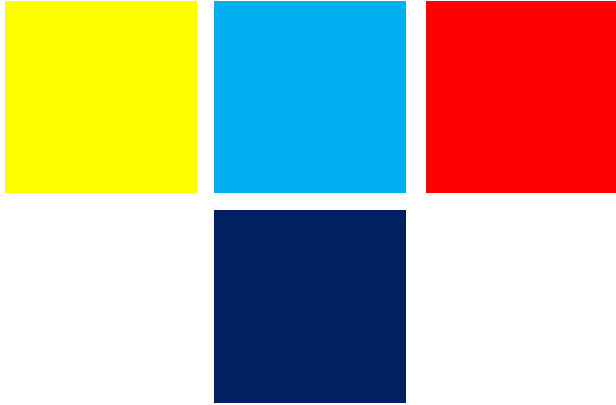
2. BRAIN

3. Knowledge creation

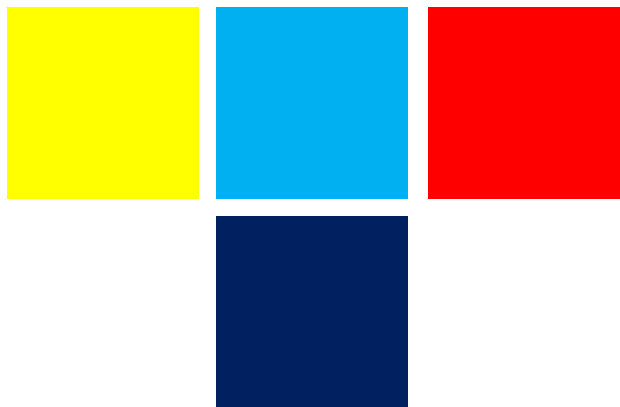
4. Results

Conclusion

References



- [1] François Chollet, “On the Measure of Intelligence”, *arXiv:1911.01547*, Nov. 5, 2019.
- [2] François Chollet, Mike Knoop, Bryan Landers, Greg Kamradt, Hansueli Jud, Walter Reade, and Addison Howard, ARC Prize website, *arcprize.org*, accessed 2025.
- [3] Brice Louvet, “We Created an AI That Thinks Like the Human Brain” (Trans.), *SciencePost*, Sep. 15, 2025.
- [4] Simon Ouellette, “Out-of-Distribution Generalization in the ARC-AGI Domain: Comparing Execution-Guided Neural Program Synthesis and Test-Time Fine-Tuning”, *arXiv:2411.17708*, Sep 21, 2025.
- [5] Julien Pourcel, Cédric Colas, and Pierre-Yves Oudeyer, “Self-Improving Language Models for Evolutionary Program Synthesis: A Case Study on ARC-AGI”, *arXiv:2507.14172*, Jul 10, 2025.
- [6] Jeremy Berman, “How I Came in First on ARC-AGI-Pub Using Sonnet 3.5 with Evolutionary Test-Time Compute”, *Jeremy’s Substack*, Dec. 6, 2024.
- [7] Daniel Franzen, Jan Disselhoff, and David Hartmann, “Product of Experts with LLMs: Boosting Performance on ARC Is a Matter of Perspective”, *arXiv:2505.07859*, Jun. 11, 2025.
- [8] Guan Wang, Jin Li, Yuhao Sun, Xing Chen, Changling Liu, Yue Wu, Meng Lu, Sen Song, and Yasin Abbasi Yadkori, “Hierarchical Reasoning Model”, *arXiv:2506.21734*, Aug. 4, 2025.
- [9] Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord, “Think You Have Solved Question Answering? Try ARC, the AI2 Reasoning Challenge”, *arXiv:1803.05457*, Mar. 14, 2018.



Thank you for listening.

Appendix

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Notifications Fork 0 Star 0

Code Issues Pull requests Actions Projects Security Insights

main 2 Branches 0 Tags

 **mercuryNS** Change grille

1ddc441 · 5 days ago 111 Commits

Arthur_1	Ajouts analyse performances	3 weeks ago
Arthur_2	Add output files for new tasks: draw_line_06, pattern_til...	last week
Data	Ajouts analyse performances	3 weeks ago
Paul	nouvelles grilles pipeline	5 days ago
Selen	Change grille	5 days ago
Timothée	Add output files for new tasks: draw_line_06, pattern_til...	last week
tasks_exam	nouvelles grilles pipeline	5 days ago
.DS_Store	Add output files for new tasks: draw_line_06, pattern_til...	last week
LICENSE	Initial commit	5 months ago
READ ME.md	Premier Commit	5 months ago

About

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0 stars

0 watching

0 forks

Report repository

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No releases published

Packages

No packages published

Contributors

 4

22